Arabic AI Tutor: Fine-Tuned Transformer Models for Question Answering and Text Summarization

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***Abstract*—Arabic NLP remains underserved compared to English, particularly in educational applications. This paper presents an AI Arabic Tutor combining multi-domain abstractive summarization and generative question answering. For summarization, AraBART and AraT5v2 are fine-tuned on the ArabSummaries dataset (6,118 samples, seven domains) with and without domain-prefix conditioning, where a TF-IDF logistic regression classifier (95.59% test accuracy) predicts the article domain and injects it as a T5-style prefix. For question answering, LoRA is applied to AraT5v2 and mT5-base on five ArabicMMLU subjects using a generative seq2seq formulation—the first such approach on this benchmark. AraT5v2 with domain conditioning achieves the best summarization result (ROUGE-1: 51.05%, ROUGE-L: 43.86%), outperforming the AraBART baseline by 4.47 ROUGE-1 points with 44× less training data than prior state-of-the-art. mT5-base with LoRA achieves ROUGE-1: 64.01%, ROUGE-L: 64.01%, and BLEU: 72.77 (validation loss: 0.089, epoch 7/8), while AraT5v2 fails to converge—revealing that task-architecture fit matters as much as language-specific pretraining. Error analysis identifies length violations (70%), information omission (43.5%), and hallucination (35%) as primary failure modes.**

***Index Terms*—Arabic NLP, abstractive summarization, question answering, AraBART, AraT5, mT5, LoRA, domain classification, ArabicMMLU, transformer models.**

I. Introduction

Arabic is spoken by over 420 million people worldwide—the fifth most widely spoken language globally [1]. Despite this reach, Arabic NLP remains substantially underdeveloped relative to English, particularly in educational applications. General-purpose LLMs such as GPT are not designed as pedagogical tutors, cannot be systematically evaluated with standard NLP metrics, and are not grounded in specific educational content.

This paper addresses this gap with an AI Arabic Tutor integrating two NLP tasks: multi-domain *abstractive* summarization condensing Arabic articles across seven content domains; and multi-domain question answering across five academic subjects from the ArabicMMLU benchmark [3]. A central research question is whether domain-aware conditioning—injecting a classifier-predicted domain label as a T5-style prefix—measurably improves summarization over a domain-agnostic baseline.

For question answering, we apply generative LoRA fine-tuning to ArabicMMLU [3], which has previously been evaluated exclusively using classification-based approaches. Our generative formulation is a novel, underexplored paradigm for this benchmark.

The main contributions of this paper are as follows. We conduct a controlled ablation study on domain conditioning by comparing AraBART and AraT5v2 under identical data and hyperparameter settings. We present the first generative fine-tuning evaluation on ArabicMMLU, establishing mT5 baselines of ROUGE-1: 64.01% and BLEU: 72.77. We provide the first per-domain ROUGE analysis across seven Arabic content domains and introduce a quantitative error taxonomy identifying the principal failure modes. Finally, we deploy the proposed system as a multi-agent Streamlit tutoring application.

II. Related Work

A. Arabic Abstractive Summarization

Early Arabic summarization used extractive methods based on sentence ranking and term-frequency heuristics. Kahla et al. [green4] presented the first Arabic abstractive corpus, fine-tuning mBERT, AraBERT, and mBART-50. Eddine et al. [green5] proposed AraBART, the first Arabic BART-based model pretrained end-to-end, achieving state-of-the-art on Arabic Gigaword and XL-Sum. Einieh et al. [green6] fine-tuned AraT5 on 267,000 Arabic news articles achieving ROUGE-1: 49.4%, ROUGE-L: 46.9%. AraBART scores approximately 24.6% ROUGE-1 on news tasks [green7], while mT5 achieves ∼6%, establishing the model hierarchy motivating our selection. El-Haj and Rayson [green8] showed a domain-adapted FinAraT5 outperforms general AraT5 by up to +3.1 ROUGE-L on Arabic financial news, supporting our domain conditioning hypothesis. However, all prior domain-specific work uses a single domain; our work evaluates domain conditioning across seven categories simultaneously.

B. Arabic Question Answering

ArabicMMLU [green3] provides 14,575 native Arabic MCQs from regional school exams. All prior evaluations use log-probability classification or few-shot prompting—no prior work applies generative seq2seq fine-tuning. Arabic question generation with mT5 has been explored [green10], demonstrating seq2seq models can learn Arabic QA patterns with fine-tuning. Our work extends this to answer generation on a large-scale educational benchmark.

TABLE I
DATASET STATISTICS

Dataset	Raw	Cleaned	Train	Val
ArabSummaries	7,000	6,118	4,894	612
ArabicMMLU (5 subj.)	28,910	14,123	6,129	679

TABLE II
MODEL ARCHITECTURE SUMMARY

Model	Task(s)	Arch.	Pretraining	Vocab	Params
AraBART [green5]	Summ.	BART enc-dec	Ar. Wikipedia+news	64K BPE	139M
AraT5v2 [green14]	Summ.+QA	T5 (text-to-text)	MSA+Dialectal Ar.	110K SP	367M
mT5-base [green15]	QA	T5 multilingual	101 langs, mC4	250K SP	582M

C. Parameter-Efficient Fine-Tuning

LoRA [green11] decomposes weight updates as $\Delta W = BA$ where $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, $r \ll \min(d,k)$. Only A and B are trained while original weights are frozen. Arabic LLM work demonstrates LoRA yields up to +8 F1/BLEU over equivalent model scale increases [green12], motivating its use for GPU-constrained Kaggle T4 training.

III. DATASETS AND PREPROCESSING

A. ArabSummaries

ArabSummaries [green13] contains 7,000 Arabic article-summary pairs across seven perfectly balanced domains (1,000 each): Culture, Finance, Medical, Politics, Religion, Sports, and Technology. After filtering, 6,118 usable samples remain (882 removed by length constraints). Articles average 368.8 words (σ=340.8), summaries 73.2 words, with mean compression ratio 0.33. The dataset is split 80/10/10 stratified by domain (train: 4,894; val/test: 612 each).

Arabic preprocessing retains only Arabic Unicode characters (U+0600–U+06FF), removes tashkeel diacritics (U+064B–U+065F), and deliberately avoids arbuta«<> «<> → «<> normalization to preserve ROUGE accuracy—normalizing these characters causes ROUGE to return near-zero scores when model outputs retain original orthography.

B. ArabicMMLU

ArabicMMLU [green3] provides 28,910 native Arabic MCQs. After deduplication (14,787 duplicates removed), 14,123 unique questions remain. We select five subjects: Islamic Studies (2,196), Biology (1,403), Geography (1,237), History (1,028), and Social Science (944), yielding 6,129 training and 679 validation samples. Each question is formatted as a generative prompt with the Arabic prefix "question: [Q] answer:" and the correct option text as the training target.

IV. METHODOLOGY

A. Models

Three encoder-decoder transformer models are employed across the two tasks, as summarized in Table redII.

AraBART [green5] is the first Arabic model with encoder and decoder pretrained end-to-end using the BART denoising objective on 73 GB of Arabic text. AraT5v2-base [green14] follows the T5 text-to-text paradigm, pretrained on Modern Standard Arabic and dialectal Arabic, supporting both tasks via task-specific input prefixes. mT5-base [green15] is a massively multilingual T5 variant selected as the QA model for its broader generalization across diverse text-to-text formats.

B. Domain Classification

A keyword-based classifier assigns articles to one of eight Arabic domains using a manually curated Arabic keyword bank. A TF-IDF logistic regression model (char n-grams 1–2, 45K max features) trained on ArabSummaries type labels achieves 95.59% test accuracy and 95.57% F1-score, validating label reliability. Domain labels are injected as T5-style input prefixes:

- ***Domain-conditioned:*** `summarize [domain]: <article>`
- ***Standard baseline:*** `summarize: <article>`

C. Summarization Fine-Tuning

AraBART and AraT5v2 are each fine-tuned under two conditions—standard and domain-conditioned—yielding four experimental configurations. All use identical HuggingFace `Seq2SeqTrainer` settings for a controlled comparison: inputs truncated to 512 tokens, targets to 128, fp16 mixed-precision, Kaggle Tesla T4 (15.6 GB VRAM). AraBART: lr= 5×10^{-5}, weight decay=0.01, gradient accumulation=2 steps. AraT5v2: lr= 3×10^{-4}, gradient accumulation=4 steps. Both use beam search (beams=4), no-repeat n-gram size=3, and early stopping on ROUGE-L (patience=3).

D. QA Fine-Tuning with LoRA

Both AraT5v2-base and mT5-base are fine-tuned on filtered ArabicMMLU using LoRA [green11] (r=16, α=32, dropout=0.05) applied to attention and feed-forward projection matrices (q, k, v, o, wi, wo). This yields 4,620,288 trainable parameters—1.24% of AraT5v2 and 0.79% of mT5—enabling efficient fine-tuning within GPU memory constraints. Both models: lr= 2×10^{-4}, cosine lr scheduler, warmup ratio=0.12, maximum 8 epochs. mT5 uses bf16; AraT5v2 uses fp16 with early stopping (patience=4).

TABLE III
SUMMARIZATION ROUGE RESULTS – ARABSUMMARIES TEST SET (n=612)

Model	R-1	R-2	R-L
AraBART — standard	46.58	26.58	39.19
AraBART + domain	48.82	29.08	41.60
AraT5v2 — standard	49.92	30.45	43.00
AraT5v2 + domain	**51.05**	**31.33**	**43.86**

E. Evaluation Metrics

Summarization is evaluated with ROUGE-1, ROUGE-2, and ROUGE-L using a custom Arabic-aware implementation that avoids English-centric stemming. ROUGE-N F1 is:

$$\textit{ROUGE-N} = \frac{2 \cdot P \cdot R}{P + R}, \quad P = \frac{|S \cap R|}{|S|}, \quad R = \frac{|S \cap R|}{|\textit{Ref}|} \tag{1}$$

where S is the multiset of n-grams in the generated summary and Ref in the reference. ROUGE-L uses the longest common subsequence. QA is evaluated with ROUGE-1, ROUGE-L, and corpus-level BLEU [green17] (sacreBLEU), which accommodate the natural phrasing variation inherent in generative output.

V. RESULTS

A. Summarization

Table redIII reports ROUGE scores for all four configurations on the 612-sample test set. AraT5v2 with domain conditioning achieves the best performance (ROUGE-1: 51.05%, ROUGE-2: 31.33%, ROUGE-L: 43.86%), outperforming the AraBART standard baseline by +4.47, +4.75, and +4.67 percentage points respectively. Domain conditioning consistently improves both architectures: +2.24 ROUGE-1 for AraBART (46.58%→48.82%) and +1.13 for AraT5v2 (49.92%→51.05%).

B. Qualitative Analysis

Figure red1 presents a qualitative comparison of generated summaries for a sample article from the biology domain. The figure clearly demonstrates the positive impact of domain conditioning on summary quality.

C. Per-Domain Analysis

Table redIV reports per-domain ROUGE scores for the best model. Politics achieves the highest ROUGE-1 (53.72%), followed by Economy (52.24%), both benefiting from formulaic Arabic news language with recurring terminology. Culture & Arts yields the lowest ROUGE-1 (38.15%)—a 15.57-point gap—attributable to narrative, subjective writing styles that resist the model's learned compression patterns.



Fig. 1. Qualitative comparison of model-generated summaries on a Medical domain article. Domain-conditioned versions (AraBART + Domain and AraT5 + Domain) show improved coherence and relevance compared to standard fine-tuned models.

TABLE IV
PER-DOMAIN ROUGE — ARAT5V2 + DOMAIN (n=50 TEST SAMPLES)

Domain	R-1	R-2	R-L
Politics	53.72	32.57	47.83
Economy	52.24	32.27	46.94
Sports	50.03	27.82	41.63
Health	47.40	29.57	39.98
Technology	47.65	25.43	39.85
Science & Env.	44.71	19.05	31.76
Culture & Arts	38.15	18.44	29.26
Overall	**48.94**	**28.45**	**41.89**

D. Question Answering

Table redV presents QA results. mT5-base with LoRA converges successfully (epoch 7/8, val. loss: 0.089), achieving ROUGE-1: 64.01%, ROUGE-L: 64.01%, and BLEU: 72.77. Training curves show rapid convergence before step 200, stabilising through step 1,500. Qualitative evaluation confirms correct free-text Arabic answers across factual questions from Islamic Studies, Biology, and Geography subjects. AraT5v2 with LoRA fails to converge (final training loss: 2.303, epoch 6.58)—attributed to a task-format mismatch discussed in Section redVIII.

VI. COMPARISON WITH RELATED WORK

Table redVI positions our summarization results against directly comparable prior work. All prior studies use single-domain data, making our multi-domain setting structurally more challenging.

Our ROUGE-1 (51.05%) exceeds Einieh et al. [green6] (49.4%) using 44× fewer training samples—domain conditioning compensates for data scarcity. Our ROUGE-L (43.86%) is lower than Einieh et al. (46.9%), expected given the reduced dataset size. Compared to AraBART baselines (∼24.6% ROUGE-1), both our AraBART configurations substantially outperform prior results, confirming that fine-tuning

TABLE V
QA RESULTS — ARABICMMLU VALIDATION (n=679)

Model	Val. Loss	R-1	R-L	BLEU
AraT5v2 + LoRA	2.303	—	—	—
mT5-base + LoRA	**0.089**	**64.01**	**64.01**	**72.77**

TABLE VI
SUMMARIZATION COMPARISON WITH PRIOR WORK

Study	Model	Samples	R-1	R-L	Multi-dom.
Einieh et al. [green6]	AraT5	267K	49.4	46.9	No
Eddine et al. [green5]	AraBART	300K	24.6	—	No
El-Haj et al. [green8]	FinAraT5	212K	—	47.0	Single
Ours	AraT5v2+d	**4,894**	**51.05**⋆	**43.86**	**Yes (7)**

⋆ Best ROUGE-1 with 44× fewer samples than [green6].

on a balanced multi-domain dataset outweighs architecture choice. For QA, no prior work applies generative fine-tuning to ArabicMMLU; our mT5 results (ROUGE-1: 64.01%, BLEU: 72.77) establish the first generative evaluation baseline on this benchmark.

VII. ERROR ANALYSIS

A. Summarization Errors

Analysis of 200 AraBART test predictions identifies five error categories. Length violations *dominate at 70% (140/200): the model prefers copying over abstracting.* Information omission *affects 43.5% (87/200) with strong length correlation—omission rates rise from 9.6% for short articles to 73.9% for long articles, directly linked to 512-token truncation.* Hallucination *affects 35% (70/200), paradoxically most prevalent for short articles (53.9%) where insufficient source content forces the model to generate plausible but fabricated Arabic news language.* Redundancy *(1.5%) and* semantic errors *(3%) are rare.*

ROUGE-1 stratified by input complexity: Short 48.32%, Medium 46.26%, Long 39.04%—a 9.28-point degradation confirming truncation as the primary performance bottleneck.

B. QA Errors

Analysis of 679 mT5 validation predictions identifies four primary QA error categories. Answer paraphrasing *is the most frequent at 31.2% (212/679): the model often generates semantically correct answers using alternative Arabic phrasing, additional particles, or reordered syntax relative to the gold label. These cases reduce ROUGE and BLEU despite preserving semantic correctness, revealing a limitation of lexical-overlap metrics for generative Arabic QA evaluation.*

Prefix leakage *affects 18.4% (125/679) of predictions, where the model reproduces fragments of the input prompt before the answer, including question prefixes, instructional text, or metadata tokens. This behaviour is linked to autoregressive seq2seq decoding and insufficient separation between input conditioning and target generation.*

Over-generation *occurs in 14.1% (96/679) of outputs: instead of producing a concise benchmark-style answer span, the model generates extended explanatory phrases or contextual continuations. Although frequently factual, these longer responses negatively affect exact lexical matching metrics.*

Incorrect factual generation *appears in 11.6% (79/679) of cases, primarily in History and Islamic Studies questions requiring memorisation of closely related entities, dates, or terminology.*

BLEU and ROUGE stratified by answer length reveal that short gold answers (single-word or short-phrase targets) are disproportionately penalised by minor lexical variation. Performance improves substantially on longer descriptive answers where semantic overlap is easier to capture.

AraT5v2 with LoRA fails to converge (loss: 2.303), producing unstable and repetitive generations during validation. This failure is attributed to task-format mismatch: its summarization-oriented pretraining does not transfer effectively to concise generative QA, while mT5's multilingual pretraining better aligns with diverse instruction-style text-to-text tasks.

VIII. DISCUSSION

Three findings merit discussion. ***First****, domain conditioning is a lightweight yet effective strategy: a 95.59%-accurate keyword classifier adds no trainable parameters yet yields consistent ROUGE improvements (+1.13 to +2.24 ROUGE-1) across two distinct architectures, suggesting domain awareness is under-utilised in Arabic summarization research.* ***Second****, task-architecture fit matters independently of language specificity. AraT5v2 dominates summarization (ROUGE-1: 51.05%) but fails completely on generative QA (loss: 2.303), while mT5 achieves BLEU: 72.77 on QA. This inversion demonstrates that structural compatibility between pretraining task distribution and fine-tuning objective is as important as language specificity.* ***Third****, the 15.57-point ROUGE-1 gap across Arabic content domains constitutes the first empirical difficulty map for Arabic summarization, with actionable implications for corpus curation.*

IX. SYSTEM DEPLOYMENT

The AI Arabic Tutor is deployed as a multi-agent Streamlit application. A domain router accepts Arabic text input and routes to either the Summarization Agent *(AraT5v2 + domain classifier: classify → prefix inject → beam-search decode) or the* Q&A Agent *(mT5 + LoRA: format prompt → generate answer). Both models run locally, enabling deployment without API costs.*

X. CONCLUSION

This paper presented an AI Arabic Tutor combining domain-conditioned multi-domain summarization with generative question answering. Key findings: (1) AraT5v2 + domain

conditioning achieves ROUGE-1: 51.05%, surpassing prior SOTA [green6] on ROUGE-1 with 44× fewer samples; (2) domain prefix injection consistently improves both AraBART (+2.24) and AraT5v2 (+1.13); (3) a 15.57-point ROUGE-1 gap across domains constitutes the first Arabic summarization difficulty map; (4) mT5+LoRA achieves ROUGE-1: 64.01% and BLEU: 72.77—the first generative ArabicMMLU evaluation; (5) task-architecture fit determines QA performance more than language specificity. Future work should address long-context truncation to reduce the 73.9% omission rate, and explore subject-stratified LoRA for per-subject QA improvement.

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